DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

Second Semester, SY 2024-2025

COURSE CODE	**:**	S-ITPC 121
COURSE TITLE	**:**	Fundamentals of Database Systems
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	None
CO-REQUISITES	**:**	S-ITPC 121LA
PROFESSOR	**:**	Ms. Azenith R. Mojica
CONSULTATION HOURS	**:**	Monday 7:00-10:00 Online Thursday: 4:00-6:00 COS 100-C Friday: 10:00-12:00 COS 100-C
CONTACT INFORMATON	**:**	armojica@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course prepares the students to acquire the skills on creating, analyzing, and designing databases. This will help increase their understanding of the essentials of database modelling and design, the languages and facilities provided by the database management systems and system implementation techniques.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. understand data models, schemas, instances, and their applications in the real world;

CLO2. design effective database schemas using Entity Relationship Diagram (ERD);

CLO3. convert conceptual model into relational schema;

CLO4. apply effective data management procedures;

CLO5. develop effective logical database design; and

CLO6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand data models, schemas, instances, and their applications in the real world.	TLO1. Effectively explain the basic concepts of databases and data models. TLO2.Explain the features of database management systems, architecture of database systems, and the role of database users. TLO3. Define the basics of the relational data model.
CLO2. Design effective database schemas using Entity Relationship Diagram (ERD).	TLO4. State reasons why many system developers believe that data modeling is the most important part of the systems development process. TLO5. Write good names and definitions for entities, relationships, and attributes. TLO6. Draw an E-R diagram to represent common business situations. TLO7. Recognize when to use supertype/subtype relationships in data modeling. TLO8. Develop a supertype/subtype hierarchy for a realistic business situation. TLO9. Develop an entity cluster to simplify presentation of an E-R diagram.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0	Week 1 Jan 20-25	• Gospel Reading, Reflection, and Prayer • Discussion of Course Syllabus **Module 0: Course Introduction** • Class Participation (MS Forms on student information and assessment of their 1st semester experience)	Schoolbook PowerPoint MS Forms	ü		2
Module 1 CLO1 TLO1 TLO2 TLO3	Week 2 Jan 27-Feb 1	• Gospel Reading, Reflection, and Prayer • Discussion **Module 1: The Database Environment and Developmental Process**	Schoolbook PowerPoint	ü		2
Module 1 CLO1 TLO1 TLO2 TLO3	Week 3 Feb 3 - 8	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (Short Quiz about Database Environment and Developmental Process)	Schoolbook		ü	2

Module 2 CLO 2 TLO4 TLO5 TLO6	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 2: The Database Approach**	Schoolbook PowerPoint		ü	2
Long Exam	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (Long Exam about the topics coved under Module 1 and Module 2)	Pen and Paper Exam	ü		2
Module 3 CLO 2 TLO7 TLO8 TLO9	Week 6 Feb 24-Mar 1	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 3: Modeling Data in the Organization**	Schoolbook PowerPoint		ü	2
Module 3 CLO 2 TLO7 TLO8 TLO9	Week 7 Mar 3-8	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (Online Quiz on Modelling Data in Organization)	Schoolbook		ü	2
	Week 8 Mar 10-15	• Gospel Reading, Reflection, and Prayer • Consultation / Completion		ü		2

Midterm Exam	Week 9 Mar 17-22	• Gospel Reading, Reflection, and Prayer • Midterm Summative Assessment on Modules 1-3	ü		2
TOTAL			10	8	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Convert conceptual model into relational schema.	TLO10. List properties of relations TLO11. Transform an E-R (or EER) diagram into a logically equivalent set of relations. TLO12. Create relational tables that incorporate entity integrity and referential integrity constraints.
CLO4. Apply effective data management procedures.	. TLO13. Explain Functional Dependency and Functional Decomposition.
CLO5. Develop effective logical database design.	TLO14. Demonstrate how relations can be normalized up to 3rd normal form. TLO15.Apply various Normalization techniques for database design improvement.
CLO6.Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self- Regulation, Real-world Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 4 CLO 3 TLO10 TLO11 TLO12	Week 10 Mar 24-29	• Gospel Reading, Reflection, and Prayer • Distribution of Midterm Grades • Sustainability Week • Discussion of **Module 4: Logical Database**	Schoolbook PowerPoint	ü		2
Module 4 CLO 3 TLO10 TLO11 TLO12	Week 11 Mar 31-Apr 5	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (Short Quiz on Logical Database) • Discussion of **Module 5: Entity Relationship Model**	Schoolbook PowerPoint	ü		2
Module 5 CLO 4 TLO13	Week 12 Apr 7-12	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (Seatwork on Entity Relationship Model)	Schoolbook PowerPoint		ü	2
	Week 13 Apr 14-19	***Independent Learning Days (April 14-15)*** ***Holy Week (April 16-19)***			ü	2

Module 6 CLO 5 TLO14 TLO15	Week 14 Apr 21-26	***Research Week*** • Pre-Discussion of **Module 6: Relational Model**			ü	2
Long Exam	Week 15 Apr 28-May 3	• Gospel Reading, Reflection, and Prayer • Enabling Assessment Long Exam about the topics coved under Module 4 and Module 5)	Schoolbook		ü	2
Module 6 CLO 5 TLO14 TLO15	Week 16 May 5-10	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 6: Relational Model** • Enabling Assessment (Short Quiz on Relational Model)	Schoolbook PowerPoint	ü		2
	Week 17 May 12-17	• Gospel Reading, Reflection, and Prayer • Consultation / Completion		ü		2
Final Exam	Week 18 May 19-24	• Gospel Reading, Reflection, and Prayer • Final Term Summative Assessment on Modules 4-6		ü		2
TOTAL				10	8	18

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.

c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.

d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Provider	Reference Material
AEA Library QA 76.9. D3. Si32 2020	Silberschatz, A. 2020. Database System Concepts. 7th ed. (New York): McGraw-Hill Education
AEA Library QA 76.9.D26 .C816 2019	Coronel, C. 2019. Database Systems: Design, Implementation and Management. 13th ed. Cengage Learning

Online References

Call Number/ E-provider	Reference Material
E-book Accession Number: **2490091**	Bush, J. 2020. Learn SQL Database Programming: Query and Manipulate Databases from Popular Relational Database Servers Using SQL. (Birmingham): Packt Publishing
E-book Accession Number: **2030800**	Lupeikiene, A. 2019. Databases and Information Systems X: Selected Papers From the Thirteenth International Baltic Conference, DB&IS 2018. (Amsterdam, Netherlands): IOS Press.
Web Resource	Introduction to Reports in Access. Microsoft 365. [accessed 2025 January 15] https://support.microsoft.com/enhttps://support.microsoft.com/en-us/office/introduction-to-reports-in-access-e0869f59-7536-4d19-8e05-7158dcd3681cus/office/introduction-to-reportshttps://support.microsoft.com/en-us/office/introduction-to-reports-in-access-e0869f59-7536-4d19-8e05-7158dcd3681cin-access-e0869f59-7536-4d19https://support.microsoft.com/en-us/office/introduction-to-reports-in-access-e0869f59-7536-4d19-8e05-7158dcd3681c8e05-7158dcd3681c

Prepared: Information Technology Department

2nd Semester 2024-2025

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